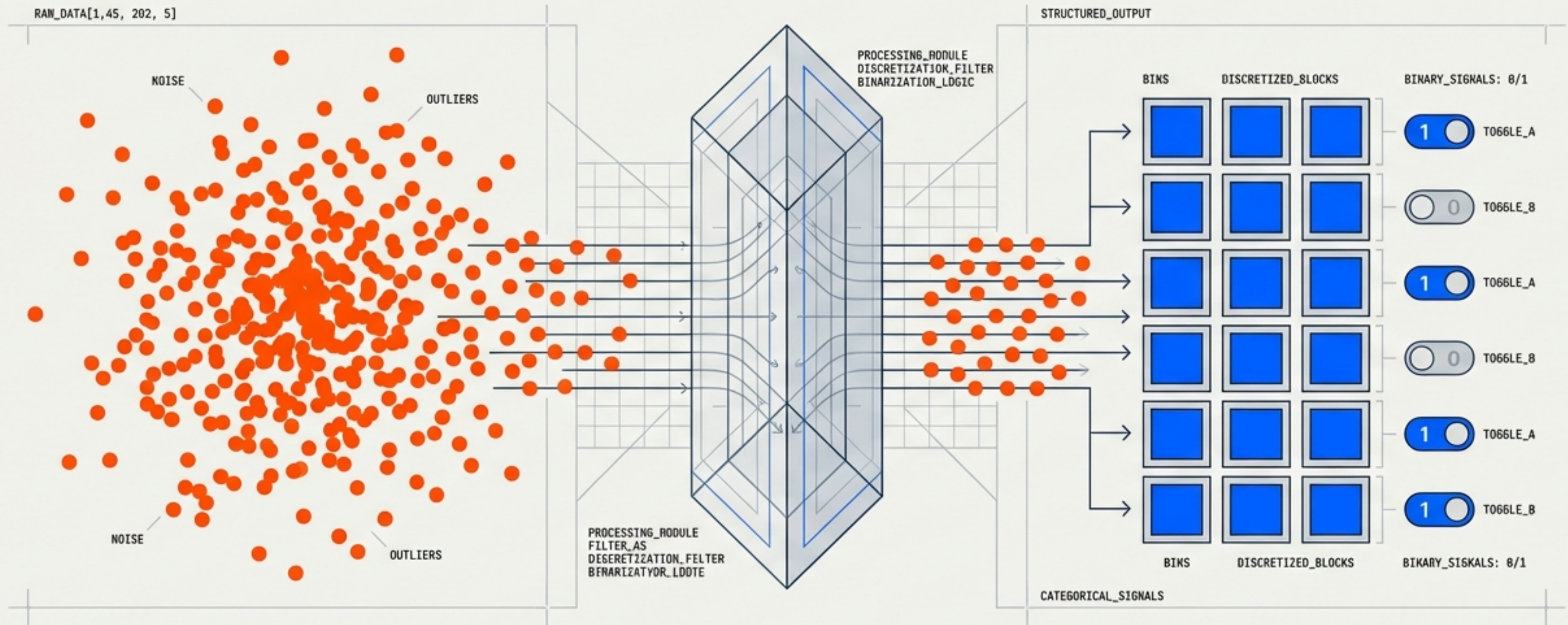


ENCODING NUMERICAL FEATURES

Mastering Discretization and Binarization in Feature Engineering



A Visual Playbook for Transforming Continuous Variables into Categorical Signals.

The Problem with Raw Continuous Data

When **precision** becomes **noise**.

In machine learning, the exact numerical difference between two data points is often irrelevant. The signal lies in the "magnitude band", not the precise decimal.

- **Goal:** Handle outliers, improve value spread, and simplify patterns for the model.

Case Study: App Downloads

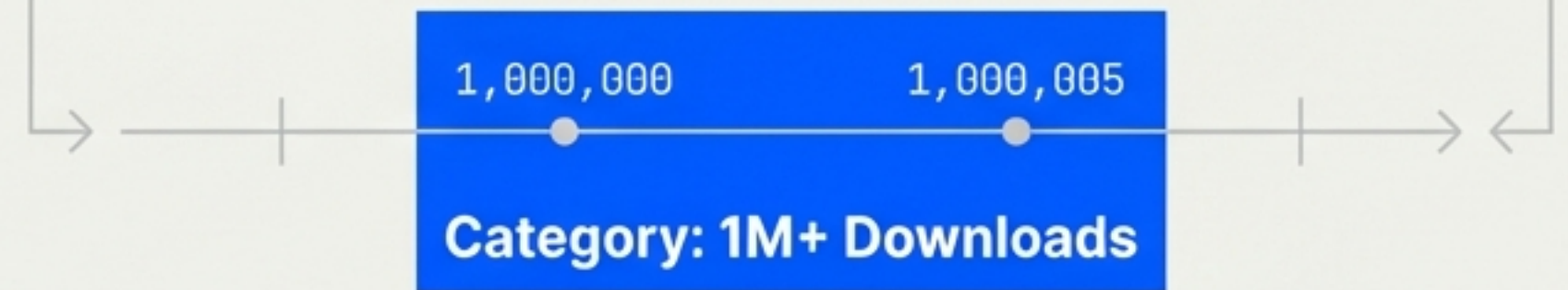
THE NOISE

Inter Medium



THE SIGNAL

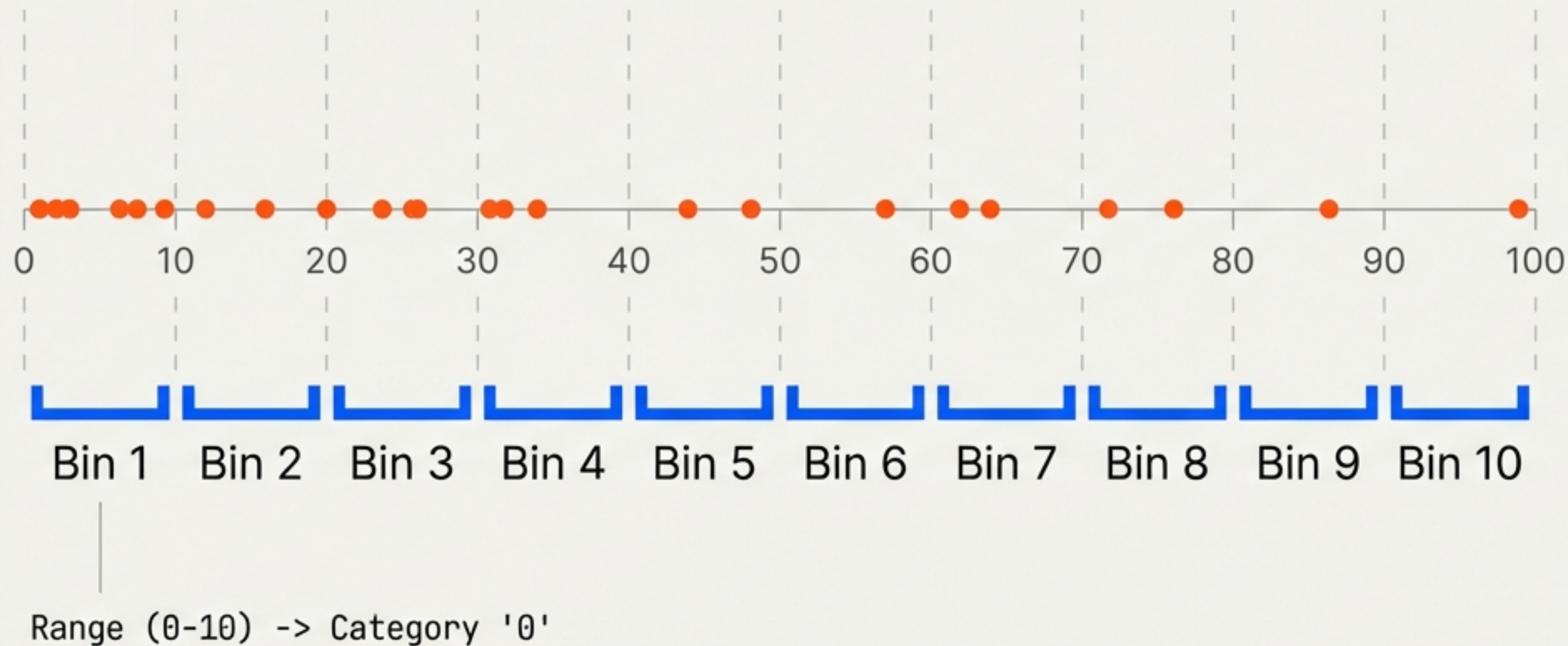
Inter Medium



Converting raw numbers into categories reduces noise and clarifies the signal.

Technique A: Discretization (Binning)

The process of transforming continuous variables into discrete intervals (bins). It functions like a histogram, mapping a range of values to a single category.



Types of Binning

1. Unsupervised
(Focus: Uniform, Quantile, K-Means)
2. Supervised
(Decision Tree Binning)
3. Custom (Domain Logic)

Strategy 1: Equal Width (Uniform) Binning

$$\text{Width} = \frac{(\text{Max} - \text{Min})}{\text{Number of Bins}}$$

Definition

Bins have the exact same width or range, regardless of data density.

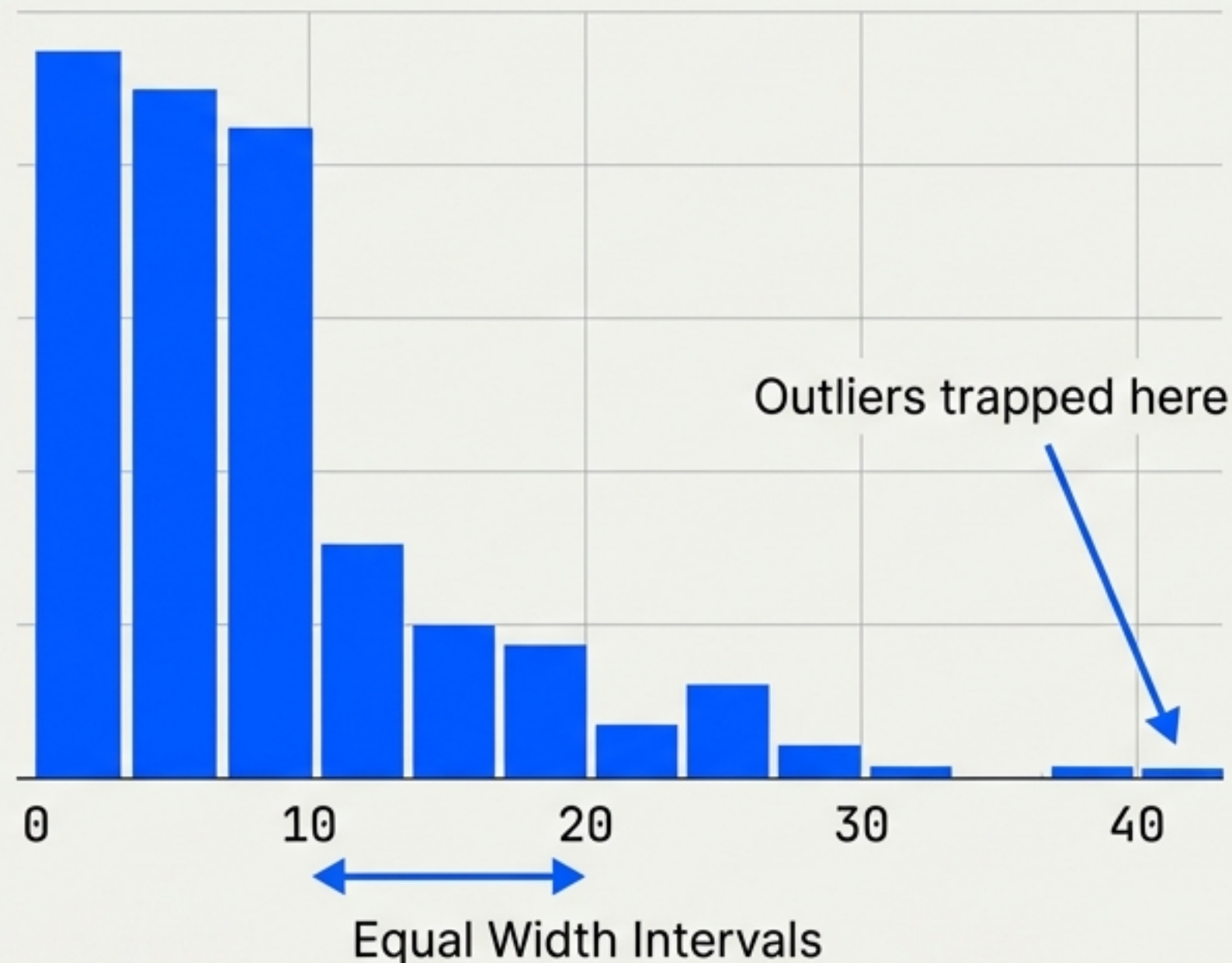
Key Insight

Great for **handling outliers**. Extreme values are trapped in the **final bin**.

Warning

Does **NOT** change the spread. Skewed data remains skewed.

Uniform Binning Distribution



Strategy 2: Equal Frequency (Quantile) Binning

Mechanism:

Uses **Percentiles**. Each bin contains the same number of observations (e.g., 10% of total).

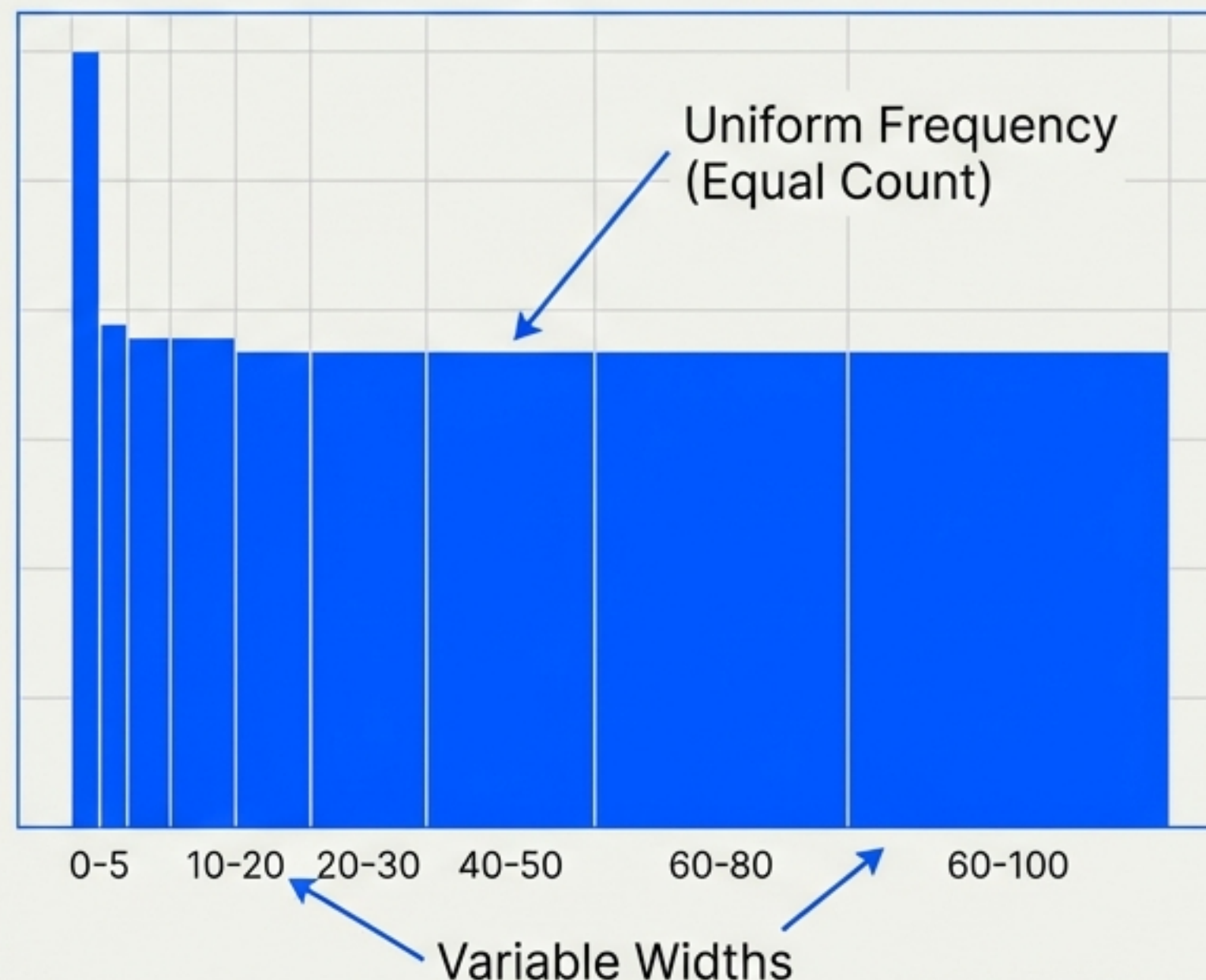
The “Default” Choice:

This is often the preferred unsupervised method.

Key Benefit:

Forces a **uniform value spread**. It flattens skewed distributions, ensuring every bin is equally represented.

Quantile Binning Distribution

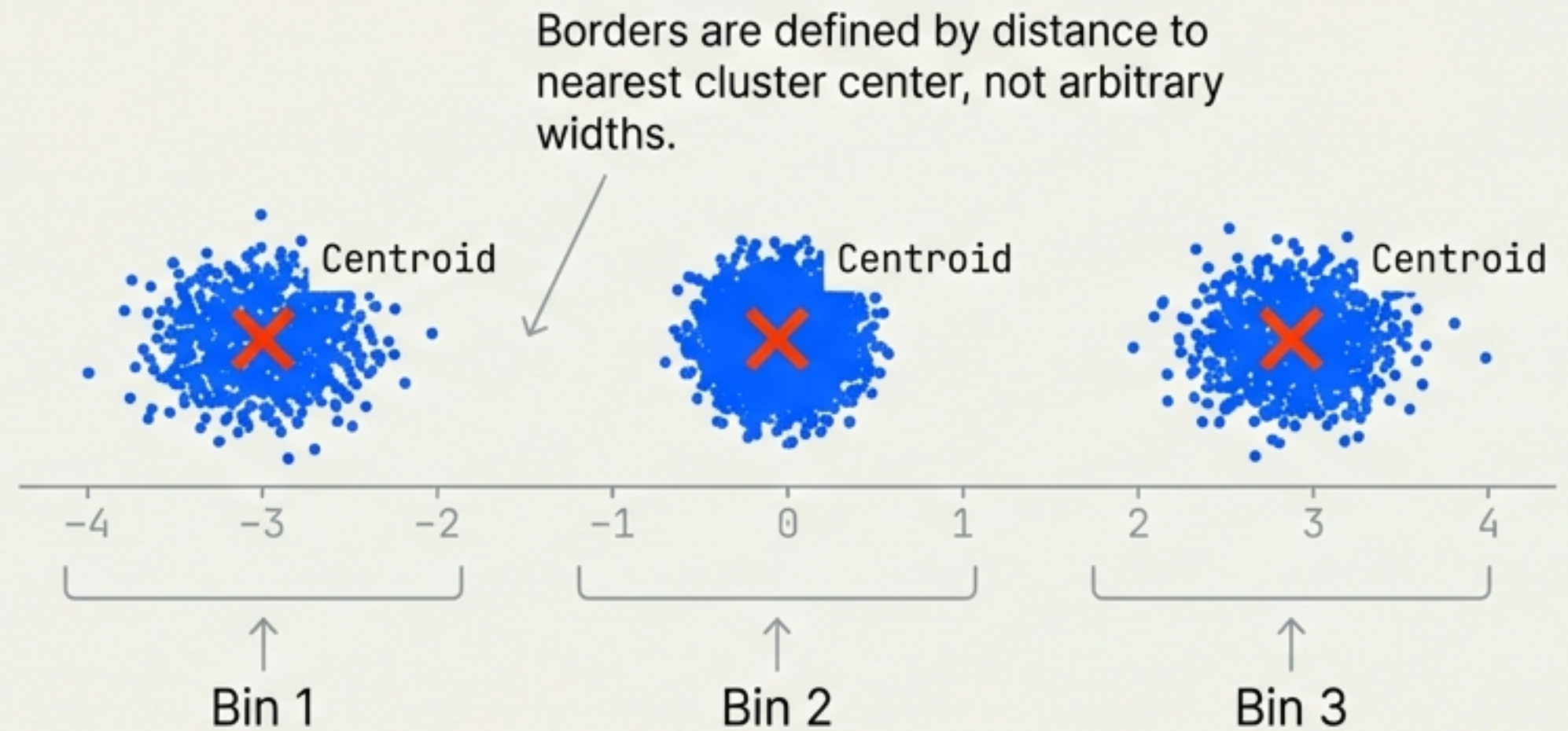


Strategy 3: K-Means Binning

Concept: Uses clustering (K-Means) to find natural groups in the data.

Logic: Identifies centroids (centers of mass). Data points are assigned to the bin of the nearest centroid.

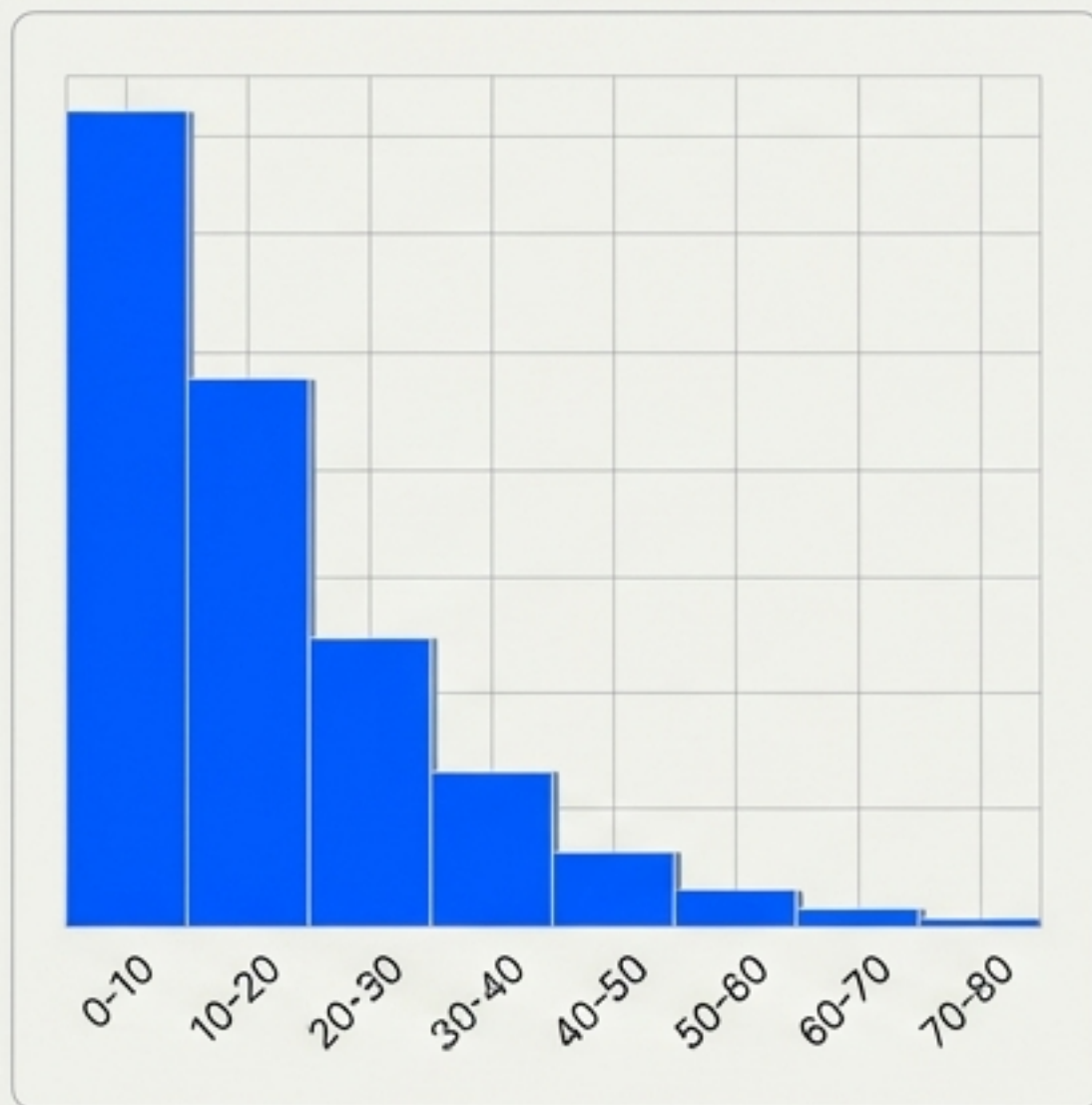
Best Use Case: When data naturally clumps together with significant gaps between groups.



Visual Impact: Comparing the Strategies

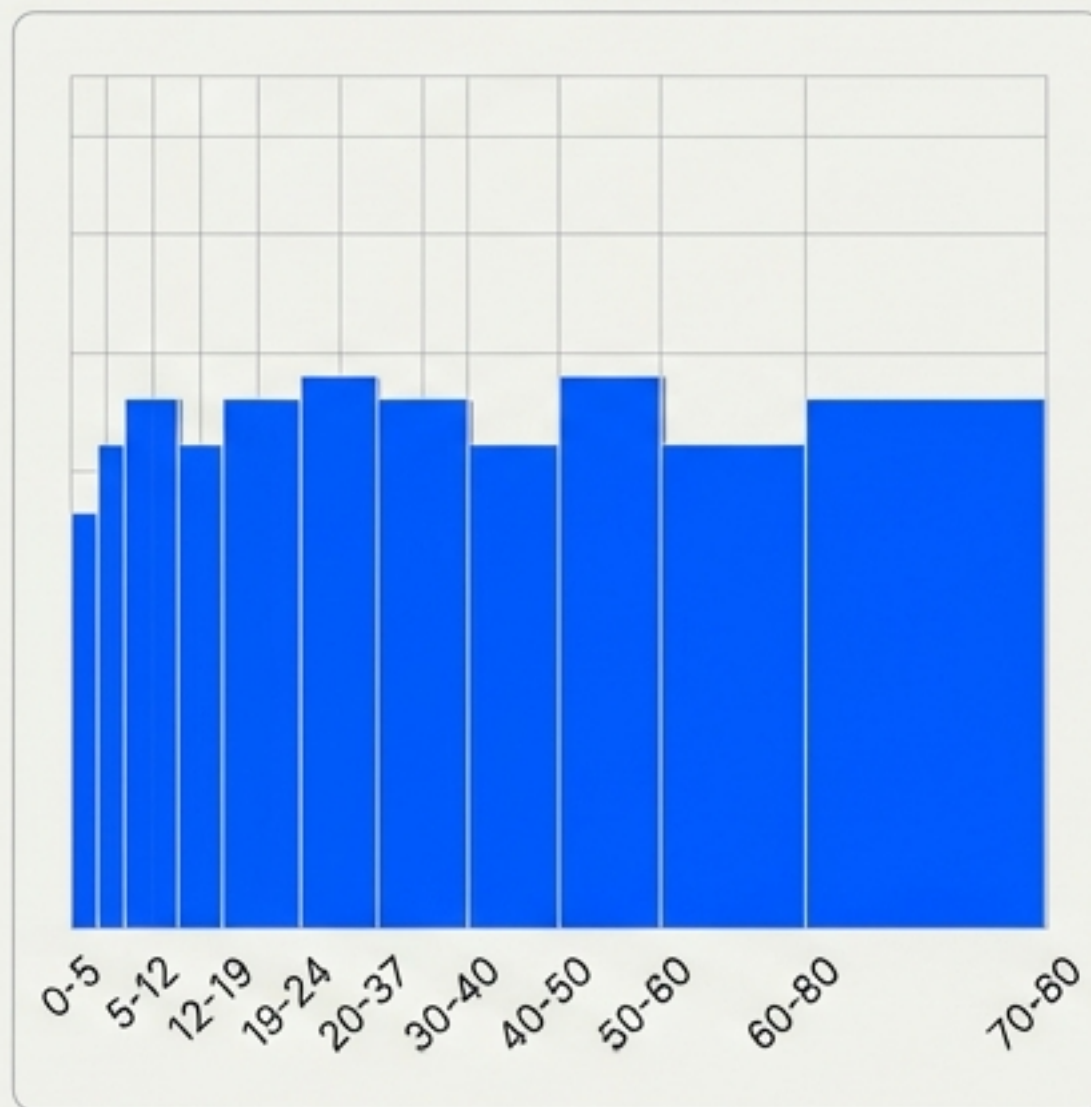
Source Data: Titanic Dataset ('Age' Column)

Uniform Binning



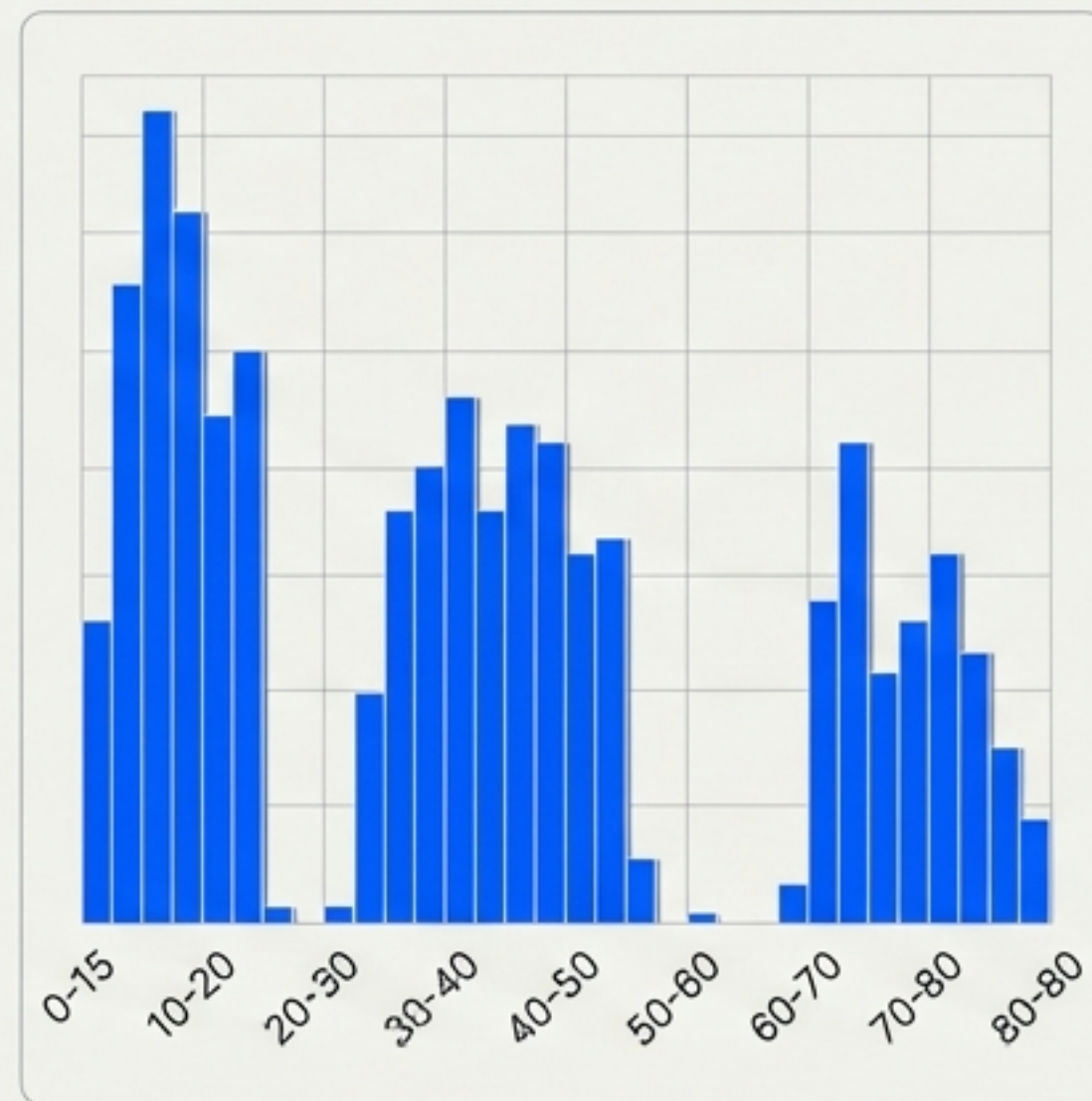
Preserves Skew / Equal Widths
Inter Regular

Quantile Binning



Flattens Distribution / Equal Count
Inter Regular

K-Means Binning



Respects Natural Clusters
Inter Regular

Implementation: Scikit-learn `KBinsDiscretizer`

Parameters Breakdown

`n\_bins`:

Number of intervals to create.

`strategy`:

- 'uniform' (Equal Width)
- 'quantile' (Equal Frequency)
- 'kmeans' (Clustering)

`encode`:

- 'ordinal' (Integer output: 0, 1, 2)
- 'onehot' (Sparse matrix output)

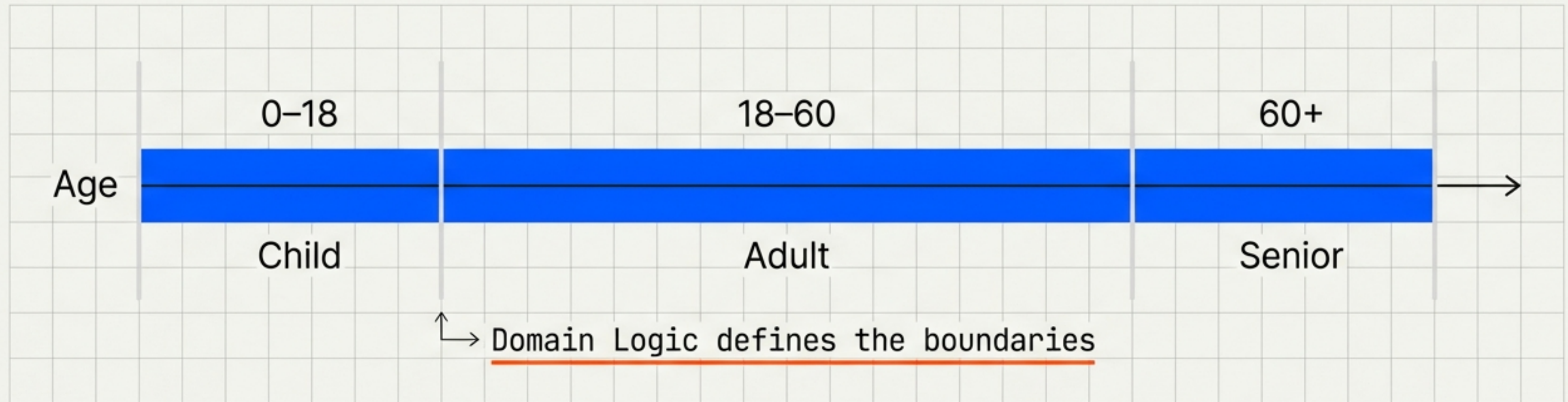
```
from sklearn.preprocessing import KBinsDiscretizer

# Initialize the Discretizer
kbins = KBinsDiscretizer(n_bins=10,
                          strategy='quantile',
                          encode='ordinal')

# Transform the data
X_binned = kbins.fit_transform(X_train)
```

The Human Element: Custom (Domain) Binning

Algorithms optimize for math; humans optimize for context. Sometimes manual bin edges based on business logic outperform algorithms.



Implementation: Requires custom logic (e.g., Pandas `cut` function), as KBinsDiscretizer does not support manual edges.

Technique B: Binarization

The Extreme Discretization

Definition:

Converting continuous numerical features into a strictly binary outcome (0 or 1).

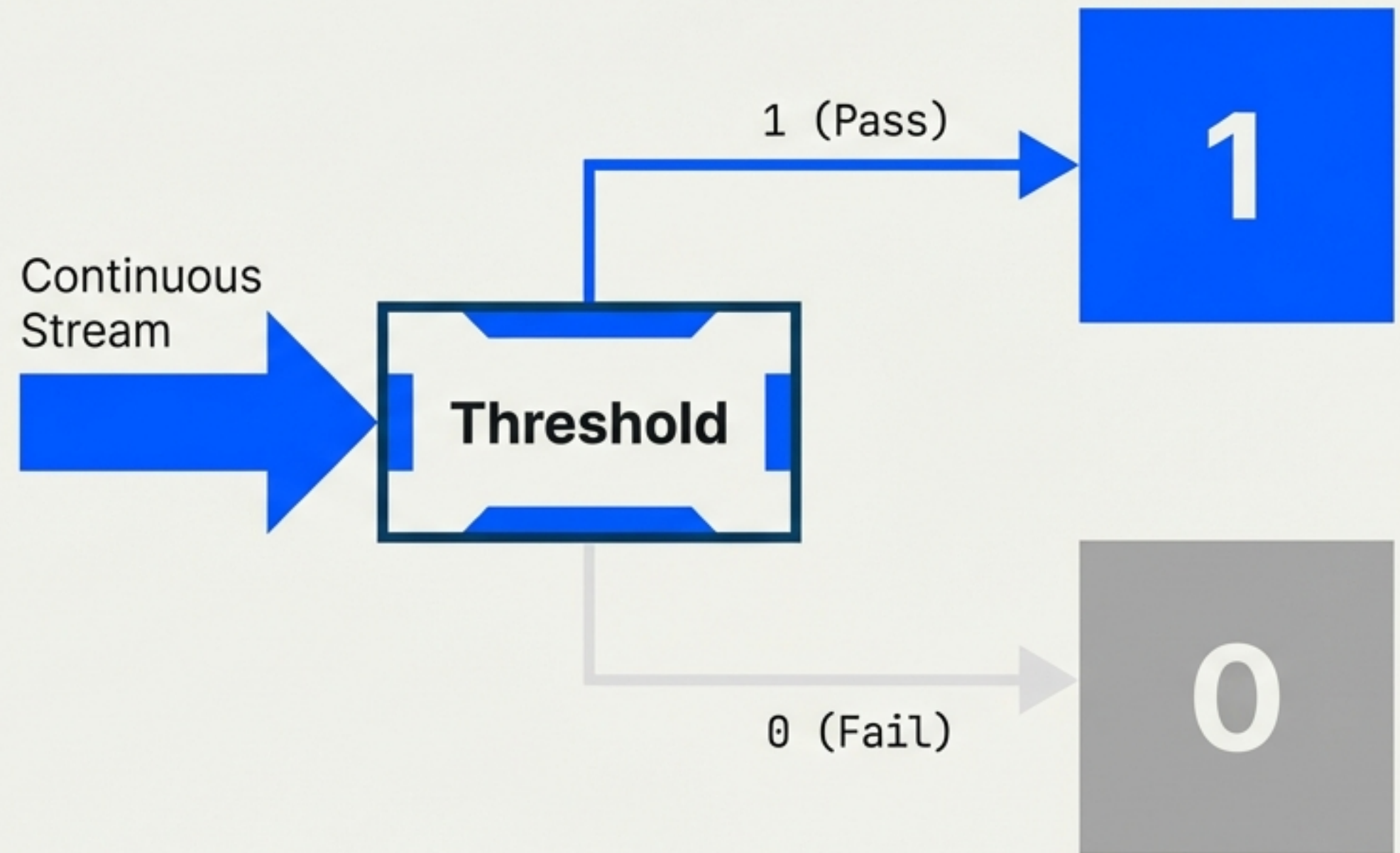
The Mechanism:

A Threshold approach. When the presence or absence of a phenomenon is more important than its magnitude.

Equation:

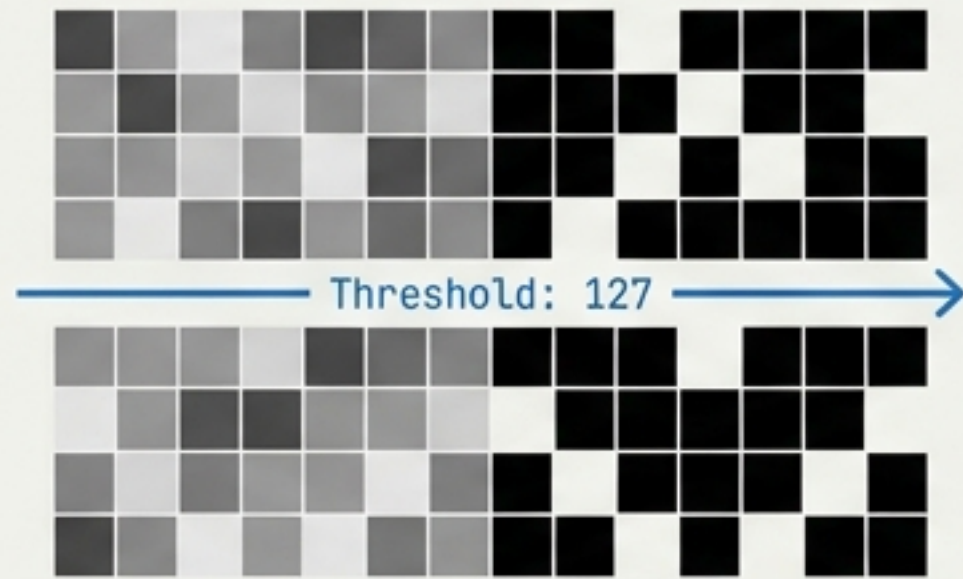
If $\text{Value} > \text{Threshold} \rightarrow 1$

If $\text{Value} \leq \text{Threshold} \rightarrow 0$



Binarization in the Real World

Image Processing



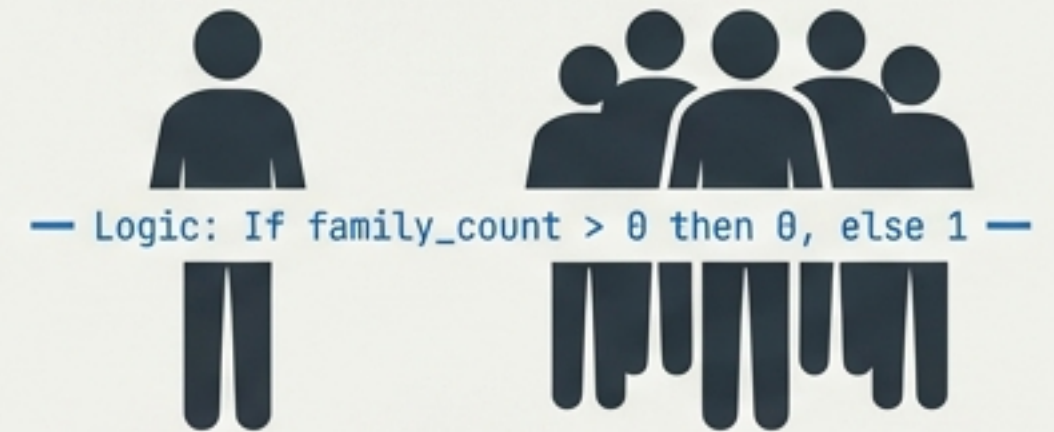
Pixels (0-255) $\rightarrow$ B&W (0 or 1).
Threshold: 127.

Finance & Law



Taxable vs. Non-Taxable Income.
Income. Threshold: 600,000.
Magnitude matters less than
status.

Feature Engineering



Titanic: "Traveling Alone".
Logic: If family_count > 0
then 0, else 1.

Implementation: Scikit-learn `Binarizer`

Intro

Class:

`sklearn.preprocessing.Binarizer`

Parameter:

threshold (The cutoff value).
Inter Regular

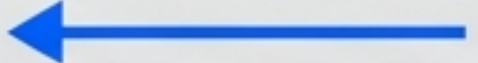
Example Logic:

Transforming Titanic 'Family' count
(SibSp + Parch) into 'Is_Alone'.

```
from sklearn.preprocessing import Binarizer

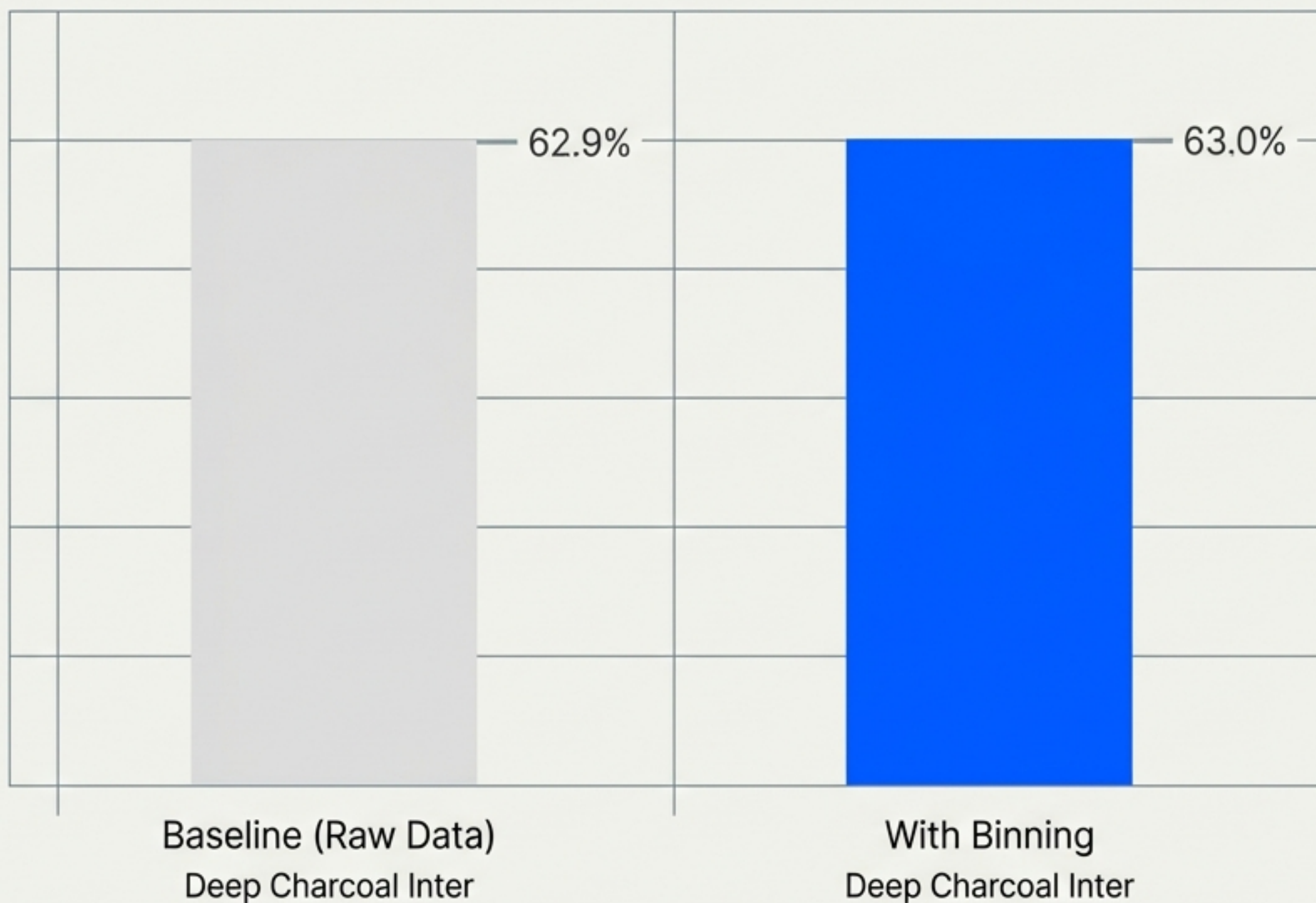
# Example: Create 'Is_Alone' feature
# Threshold = 0 (If >0, has family. If <=0, alone)

binarizer = Binarizer(threshold=0, copy=True)
is_not_alone = binarizer.fit_transform(X[['Family_Size']])
```



Everything above
0 becomes 1

The Reality Check: Performance



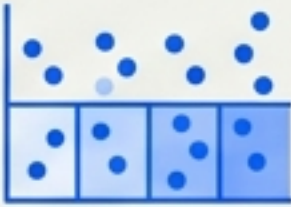
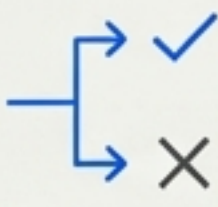
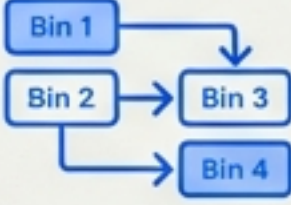
The Insight

Feature engineering provides robustness and noise reduction, but it isn't magic. In the Titanic experiment, binning prevented overfitting but didn't drastically change accuracy.

Takeaway

Meaningful jumps depend on data nature. Binning is a tool for stability, not always a silver bullet for accuracy.

Summary: Which Strategy to Choose?

Objective/Scenario	Recommended Strategy	
Need to handle outliers but preserve shape?	Uniform Binning	
Need a uniform value spread?	Quantile Binning (Default)	
Data has natural clusters/gaps?	K-Means Binning	
Need a simple Yes/No flag?	Binarization	
Have specific business rules?	Custom/Manual Binning	

“Binning turns specific, noisy numbers into broader, robust signals.”